

;===== Device Structure=====

```
(sdegeo:create-rectangle (position -0.15 0 0) (position 0.15 0.2 0) "Silicon"
"r.substrate")
```

```
(sdegeo:create-rectangle (position -0.15 0 0) (position -0.1 0.05 0) "Silicon"
"r.source")
```

```
(sdegeo:create-rectangle (position 0.15 0 0) (position 0.1 0.05 0) "Silicon"
"r.drain")
```

```
(sdegeo:create-rectangle (position -0.1 0 0) (position 0.1 -0.01 0) "SiO2"
"r.oxide")
```

;===== Contacts =====

```
(sdegeo:define-contact-set "gate" 4 (color:rgb 1 0 0) "###")
```

```
(sdegeo:define-contact-set "source" 4 (color:rgb 0 1 0) "###")
```

```
(sdegeo:define-contact-set "drain" 4 (color:rgb 0 0 1) "###")
```

```
(sdegeo:define-contact-set "substrate" 4 (color:rgb 1 1 0) "###")
```

```
(sdegeo:set-contact (list (car (find-edge-id (position 0 -0.01 0)))) "gate")
```

```
(sdegeo:set-contact (list (car (find-edge-id (position -0.125 0 0)))) "source")
```

```
(sdegeo:set-contact (list (car (find-edge-id (position 0.125 0 0)))) "drain")
```

```
(sdegeo:set-contact (list (car (find-edge-id (position 0 0.2 0)))) "substrate")
```

;===== Doping =====

```
(sdedr:define-constant-profile "ConstantProfileDefinition_1" "BoronActiveConcentration" 1e16)
```

```
(sdedr:define-constant-profile-region "ConstantProfilePlacement_Substate"  
"ConstantProfileDefinition_1"  
"r.substrate")
```

```
(sdedr:define-constant-profile "ConstantProfileDefinition_2" "ArsenicActiveConcentration" 1e18)
```

```
(sdedr:define-constant-profile-region "ConstantProfilePlacement_drain" "ConstantProfileDefinition_2"  
"r.drain")
```

```
(sdedr:define-constant-profile "ConstantProfileDefinition_3" "ArsenicActiveConcentration" 1e18)
```

```
(sdedr:define-constant-profile-region "ConstantProfilePlacement_Source" "ConstantProfileDefinition_3"  
"r.source")
```

```
;===== RefEval Window =====
```

```
(sdedr:define-refeval-window "RefEvalWin_bulk" "Rectangle"  
  (position -0.15 -0.01 0)  
  (position 0.15 0.2 0))
```

```
(sdedr:define-refeval-window "RefEvalWin_channel" "Rectangle"  
  (position -0.1 0 0)  
  (position 0.1 0.02 0))
```

```
;===== Meshing =====
```

```
(sdedr:define-refinement-size "RefinementDefinition_1"  
  0.05 0.05 0.0 0.01 0.01 0.0)
```

```
(sdedr:define-refinement-placement "RefinementPlacement_1"  
  "RefinementDefinition_1"  
  (list "window" "RefEvalWin_bulk"))
```

```
(sdedr:define-refinement-size "RefinementDefinition_2"  
  0.005 0.01 0.0 0.001 0.005 0.0)
```

```
(sdedr:define-refinement-placement "RefinementPlacement_2"  
  "RefinementDefinition_2"  
  (list "window" "RefEvalWin_channel"))
```

```
(sde:build-mesh "" "n@node@")
```